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Design for repair: the mud mosque that is rebuilt every year

The free guide to the reel on the Great Mosque of Djenné: why its wall is replastered every year, how the palm sticks became a built-in ladder, and how to design any building so that repair is easy.



How it works

The Great Mosque is built of earth. Earth is strong when dry and weak in rain, so the outside wall is covered with a plaster of earth and water that the rain slowly washes away.

Instead of fighting this, the town plans for it. Every year, before the rainy season, people climb the building and put a fresh layer of mud on it. Palm sticks called toron, about 60 cm long, stick out of the walls and work as scaffolding, so none has to be brought in.

The repair is part of the design: the ladder is built into the wall, the material is local, and the whole community knows how to do the work.

Key facts

The current mosque was rebuilt around 1907 (sources say 1906-07, some up to 1909). It is often called the largest mud-brick building in the world. The annual replastering, the crépissage, is coordinated by a local guild of masons and is also a community festival.

The site has been on UNESCO's World Heritage in Danger list since 2016.

Built examples

Great Mosque of Djenné, Mali (1907)	Mud walls with toron palm sticks, replastered yearly by the community.
Djingareyber Mosque, Timbuktu, Mali	Restored by the Aga Khan Trust for Culture (2007-10) with local masons: watertight roof, consolidated mud masonry.
Alcázar of King Don Pedro I, Carmona, Spain	A rammed-earth structure studied for damage and risk to plan repairs.
Rammed-earth houses, Albaicín, Granada, Spain	A recovery plan protects 17th-century earth buildings.

Try it yourself

1. Make two small bricks of clay or garden soil mixed with water and a little straw; let them dry.

2. Cover one brick with a thin plaster of the same mix and leave the other bare.
3. Put both under a watering can for one minute, then compare the damage.
4. Repeat the test after replastering the first brick with a fresh thin layer.
5. Sketch your own building and mark every part that will wear out first; decide who repairs it, how often, and how they reach it.
6. Add a simple access feature (a ledge, a hook or a rung) to your drawing so that repair needs no special equipment.

The rule to remember

Next time you design for harsh weather, plan the repair on day one.

Hiding the weak point	Show the parts that wear, so people can fix them.
Needing special tools	Use materials and methods the local community already knows.
No repair schedule	Write down who repairs what, and when.

Go further

Djenné Great Mosque Restoration, Archnet
https://archnet.org/sites/6395

Cultural development in Mali, Aga Khan Development Network
https://www.akdn.org/where-we-work/west-africa/mali/cultural-development-mali

Mud Architecture: The Great Mosque of Djenné, ArchEyes
https://archeyes.com/great-mud-architecture-mali-dogon-culture/

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